Infusing AI with Brain Anatomy: Modelling Multisensory Integration with Cortically-Embedded Recurrent Neural Networks

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Declaration

I hereby declare that the work presented in this report is my own original work and has not been submitted, either in the same or different form, for a degree at this or any other university.

All information derived from the work of others has been acknowledged in the text and references are given in the bibliography.

AI has been used only for spell checking, latex formatting and minor coding computations such as formatting plots.

This project did not require ethical review, as determined by my supervisor Dr Seán Froudist-Walsh.

Signature: Guy Davies   Date: 28/04/2025

Abstract

The human brain continuously processes multiple sensory inputs, filtering and transforming them to enable complex cognitive functions. Creating accurate models of this could accelerate development in multimodal AI and robotics, allowing different modalities to be merged into a single coherent experience. Current neuroimaging technology cannot fully explain these cognitive mechanisms; recording tools are either high spatial (fMRI) or high temporal (MEG) resolution, but not both. Computational models lack the anatomical relevance required to explain how integration occurs in the human brain, preventing neuroscientists from making testable predictions. Conversely, cortex-wide models which incorporate biological data are often unable to perform complex cognitive tasks, limiting their use. I propose cortically-embedded recurrent neural networks (CERNNs) to bridge this gap, incorporating anatomical data in RNN models to encourage the natural emergence of brain-like behaviour. CERNNs are simultaneously trained on a variety of unisensory and multisensory tasks, and exhibit a range of biologically realistic connectivity and activity patterns. Core aspects of multisensory integration and processing can be replicated by tuning anatomically-inspired constraints, and remarkable similarities are found between CERNN and fMRI activity. CERNNs can give experimental neuroscientists a new type of tool for making informed predictions about the brain. Cognitive mechanisms identified by these models can be applied in creating artificial multisensory models, contributing to the wider co-development of neuroscience and machine learning.

Contents

1 Introduction and Literature Review

1.1 Introduction

Humans constantly perceive the world around them, and tasks often require the use of multiple modalities to complete effectively. Different brain areas are highly specialised and arranged in a functional hierarchy, with more generalisation as hierarchical depth increases.[1] When sensory inputs from multiple modalities are needed to complete a cognitive task, it is not always clear how this is carried out in the brain. Inputs need to be integrated but the location and dynamics of this are difficult to measure in humans, and current models cannot accurately predict this behaviour.

Recurrent neural networks (RNNs) have been shown to be successful in modelling complex human-like behaviour, exhibiting a variety of brain-like behaviours.[2–5] Powerful computational techniques such as backpropagation make them a capable and versatile tool for completing more complex tasks than previous models. However in neuroscience, neural network models have previously acted as an analogy of the brain rather than an *in silico* experimental model system. They lack a physical aspect necessary to accurately model spatial dynamics in the brain. This limits them to modelling neural mechanisms more abstractly, rather than drawing direct comparisons with biological systems.

Incorporating anatomy can bridge this gap, but relies on biologically realistic constraints, such as wiring costs, dendritic spine counts and multimodal inputting, in order to better emulate the brain. Constraining models in this way is more likely to develop brain-like solutions to cognitive problems, and therefore could offer a new way of forming experimental insights and predictions for neuroscientists.

The brain's organisation is commonly thought of as being a one-dimensional hierarchy, where higher regions are capable of more information processing.[6] However when a single sense is considered, it is apparent that each primary sensory region must have its own independent hierarchy. Information follows separated paths, but must converge in multimodal areas or networks. The location of these areas is not fully understood, and it is not known how or where this integration takes place. Moreover, the extent to which task context affects this integration spatially and temporally is unclear. For a model to give comprehensive results, a range of tasks must be applied.

Here, I propose cortically-embedded recurrent neural networks (CERNNs) to gain further insights into the workings of the brain, specifically its spatial optimisation and organisation. These models use anatomical data to infuse RNNs with cortical locations. Realistic solutions are encouraged by biologically-inspired constraints in the cost function. Models were simultaneously trained on 9-15 tasks requiring unisensory and multisensory processing.

CERNNs exhibit a range of biologically-realistic behaviour in connectivity and activity patterns. Changing task paradigm and complexity gives different activity, facilitating direct comparison to fMRI neuroimaging data. This comparison is impactful for neuroscience because models can be verified and once trusted, model predictions can be used in guiding experiments or even treatment in the brain. While a vast and comprehensive set of tasks would be needed to develop an accurate and flexible whole-cortex model, by selecting a small set of tasks which covers key cognitive areas, such as working memory, response inhibition and decision making, core cognitive mechanisms can emerge.

1.2 Neural Networks as Models of the Brain

Artificial neural networks (ANNs) are powerful tools in data science and machine learning. Having been originally inspired by the brain and neuronal function,[7] they are suitable for modelling a wide range of systems and functions in the brain. Impressive results across a wide range of applications in neuroscience have been achieved by using ANNs, often improving on previous models.[2] They excel at this due to their ability to learn complex relationships difficult to model with traditional techniques.[8] However there are also problems with using ANNs, mostly regarding biological implausibility. The most common learning algorithms for ANNs are simply not biologically realistic as they rely on features like two phase learning and symmetric synapses. Advances have been made to solve these issues and create more plausible models,[9] but they have not yet been widely adopted within the field. Though the way these models learn might not be realistic, the results found and conclusions drawn from them can still be valid and insightful.

Image classification is a problem which drove the development of convolutional neural networks (CNNs). Consequently, deep CNNs have become extremely proficient in the field of computer vision, also being directly inspired by neuroscience.[10] These breakthroughs provoked similar research into modelling human vision using deep learning. Yamins and DiCarlo showed that hierarchical CNNs have the potential to learn intricate details of facial expressions, suggesting functional similarities to biological vision systems.[11] Their influential paper also showed that deep layers of CNNs can explain variance in neural activity of wider brain areas. This had previously been out of the scope of computational models due to the regions' distance from sensory inputs. While the layers of a CNN could be compared with the cortical hierarchy seen in the brain, others have criticised the use of deep learning as a model of the brain. Bowers et al. highlighted key problems such as lack of generalisation, oversensitivity to surface features and biological implausibility,[12] the last of which being a recurring issue with neural network models.

To model many aspects of cognition, it is necessary to include methods of memory retention, or at least some kind of temporal dynamics. Recurrent neural networks are proving useful in more and more sectors of computational neuroscience as their architecture includes feedback loops which facilitate these time-dependencies. Many aspects of neuroscience, like excitatory-inhibitory networks,[13] reward-based learning,[3] and biologically plausible learning rules[14]have been successfully modelled using RNNs, demonstating their suitability for modelling the brain.

Despite these impressive innovations in the field, almost all representations of recurrent neural networks had no physical relevance to the anatomy and organisation of the brain and its constituent areas. Achterberg et al. proposed spatially-embedded recurrent neural networks (seRNNs), which arrange units throughout a cube and penalise long distance connections as part of regularisation, similar to how the brain does to increase metabolic efficiency.[4] When trained, these networks self-organised, displaying brain-like features such as small-worldness and mixed-selective coding. Solutions to tasks were found by developing modular groupings of nodes with shared or similar functions. This type of anatomical constraint could encourage optimisation in similar ways to the brain, and even explain how its intricate and complex structure has evolved. The development of biological characteristics in neural networks due to physical conditions marks a significant advancement in understanding the brain through computational models.

Another challenge when adapting neural networks to the brain is its remarkable ability to perform multiple cognitive tasks flexibly. Traditional experimental and modelling approaches often only include single tasks, which limits understanding of the underlying mechanisms for multitasking. Yang et al. focused on this by training single RNNs to perform a variety of tasks and related this to neural representations in the brain.[8] The authors chose 20 tasks commonly seen in cognitive testing which involved key paradigms such as working memory and decision making. The study found that learning led to compositionality in task representations, meaning that tasks could be performed by recombining instructions from other tasks. This compositionality is essential for cognitive flexibility, allowing the brain to adaptively switch between tasks. Although the network of Yang et al. did not contain any anatomical constraints, the functional clustering of units may resemble how the brain contains functional subnetworks that specialise in distinct aspects of cognition.[15] Yang's paper provides a framework for task-switching and generalisation, which will be expanded on in this project.

1.3 Brain Compositionality

Despite changing in specifics over time, studies have highlighted the modularity and compositionality of the brain for a long time. Ferrier was one of the first to discover this, demonstrating that different parts of the brain accounted for different functions, specifically within the motor cortex.[16] This evidence supported the idea that the brain is not a homogeneous structure but consists of distinct specialised regions.

Modern neuroscience offers a deeper knowledge of the brain's regions and their respective functions. An influential review from Miller and Cohen showed the prefrontal cortex (PFC) to be a complex and interconnected region responsible for a wide range of higher cognition.[17] They showed that different subdivisions of the PFC have unique but overlapping connectivity patterns, simultaneously allowing for local specialisation and integration of information from various distributed sources.

Another important discovery from the paper is the way multiple modalities are integrated and processed in the PFC in order to complete multisensory tasks. Parts of the PFC receive inputs from the visual, auditory and somatosensory cortices, and so must be multimodal to combine these simultaneously. This is crucial for representing tasks and activity across different types of information, which is necessary for adaptive behaviour. These results will play a large role in this project, which looks specifically at how different sensory inputs go from their respective cortices to being integrated in order to solve a task.

With so many brain areas, a mapping is necessary to analyse activity or create a model. Numerous such mappings, or parcellations, have been created to split the brain into named regions. The methods for which vary, and each has a different utility based on its intended use.[18] Glasser et al. used data from the Human Connectome Project to create a multimodal parcellation of the human cortex which is split into 180 areas per hemisphere based on architecture, function, connectivity and/or topography.[19] This parcellation will be used in this project as it gives a detailed structure to explore brain function and regional specificity. Note that for other applications, a different parcellation would be preferred. For example, when looking at the dynamics of the brain or the speed at which information is transmitted, a more spatially-focused or connectivity-based parcellation could be more effective.

1.4 Multisensory Integration

Efforts across all cognitive sciences have made great progress in explaining the processes underlying multisensory integration, however many aspects this complex behaviour are not yet understood. Neuroimaging has shown many regions are used in combining visual, auditory and somatosensory information, although the circumstances under which each area is used are unclear. For example, areas in the posterior temporal lobe, specifically the superior temporal sulcus, are likely to be responsible for a lot of audio-visual integration,[20] and more generally, areas in the posterior parietal lobe and the lateral occipitotemporal junction are often shown to facilitate bimodal and trimodal integration processing.[21–23] These regions are to be expected in a way, as they mostly lie in the space between the primary sensory cortices V1, S1 and A1. This also suggests that the brain is functionally organised to combine sensory inputs in early regions of the hierarchy. This information can then be sent forward to be used in the transmodal associative regions responsible for higher cognitive processing.

Experimentalists across all fields of cognitive science have investigated the effect of multisensory integration on perception, and have found that we reach a more accurate perception by combining cues from different senses and weighting them based on their respective reliabilities.[24,25] With this is mind, it is plausible that integration happens early in the hierarchy in order for the most reliable perception of reality to be used in cognitive processing.

Frameworks based on fMRI data have been developed to better understand the way associative regions high in the cortical hierarchy seem to be transmodal, and primary sensory regions are unimodal. Wei et al. emphasise the need for sensory integration for higher cognitive function, and detail a unimodal to transmodal gradient along the hierarchy with abstract areas having more integrated sensory representations.[26] Interestingly, they also base their analysis on the anchoring role which primary sensory regions play in the spatial arrangement of regions in the brain, as supported by a long line of research.[1,27,28]The foundational role that sensory regions have in the brain suggests that a model based around sensory integration could cause brain-like organisation to emerge, giving further insights into neurodevelopment. Sensory regions, particularly the visual cortex, take up large specialised sections of the brain. This could be because a sophisticated sensory system was required in order to develop the complex cognitive capabilities which humans possess. If so, these capabilities could be present in models which first successfully match the sensory processes of the brain.

The difficulty in finding the specific mechanisms of multisensory integration is likely due to the fact it is not a single process for all types of input. Functional MRI (fMRI) studies suggest a mutual interaction across multiple, often distributed, regions is necessary.[29,30] It is therefore probable that there is some sort of distributed network for combining sensory input, as is the case for many other aspects of cognition.[31] Despite this, the constituent areas of this network vary from study to study, and are seemingly sensitive to changes in the type of information, the combination of modalities and the paradigm of the experiment.[22]

Finally, neural networks have been used to model multisensory integration,[32–34] particularly with applications in robotics.[35] These studies achieve success in creating a perception of similar accuracy to a human, but without anatomical data we cannot expect them to do this in a brain-like way. Neuroscience has improved aspects of perception in robotics,[36] and a better understanding of how the brain combines modalities to gain a better overall world-view could be impactful in replicating this artificially, potentially capturing the flexibility of the brain.

1.5 Project Aims

Models of multisensory integration do not currently help in understanding how it occurs in the brain. This report aims to develop a framework which can be trained on suitable tasks with the goal of solving them in realistic ways, exhibiting brain-like behaviour. Solutions will be assessed on several key factors: task performance, learned connectivity, spatial activity and processing regions, and temporal dynamics and flow. To determine whether these properties are brain-like, qualitative and quantitative analysis can be carried out, as well as comparison with fMRI data.

2 CERNN Model

The core framework used here to create multisensory models is the Cortically-Embedded Recurrent Neural Network (CERNN), which I have developed with the the Cognition, Anatomy and Neural Networks (CANN) lab at the University of Bristol. To be clear, members of the lab had been working on this for several months before I began my project, and major parts of the model structure had already been created. However, no key results had been achieved, the model was unimodal, and major bugs prevented its effective use. Over the course of my project I worked with the lab, contributing to improve many aspects of the core framework, however modelling multisensory integration was my personal project and this is exclusively my own work.

2.1 Model Architecture

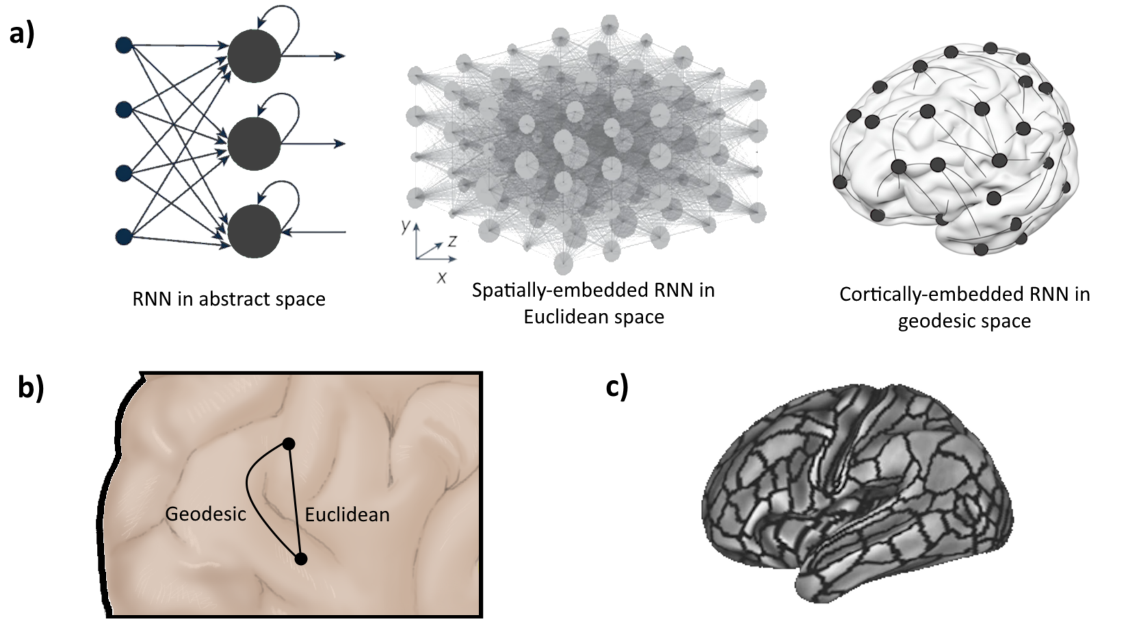

Figure 1: a) Schematics for various types of RNN are shown. A standard RNN in abstract space has node structure but no spatial interpretation. Spatially-embedded RNNs (seRNN)[4] can be used to penalise distant connections, but are not suitable for modelling the brain. Cortically-embedded RNNs (CERNN) have nodes arranged on the surface of the brain and distances are calculated on this surface. b) Euclidean distance takes the shortest path, however geodesic distance accounts for the topology of the brain, improving bio-realism. c) The Glasser parcellation[19] has 180 regions per hemisphere. In the CERNN model, each of these regions is represented as a unit.

The structure of CERNNs expands on the framework provided by spatially-embedded RNNs discussed earlier[4] by incorporating anatomical data. Rather than being arranged uniformly in Euclidean space, units are distributed across the surface of the brain, as shown in Figure 1. The locations of units are given by the Glasser parcellation[19] and the geodesic distances between each pair of units is calculated and stored in a connectivity matrix. Distance between regions is defined as the average distance of all connections between points in the two regions. During training, these values are used as in a spatial regulariser which penalises long distance connections, favouring units representing nearby brain areas. This is intended to capture the biological cost of creating and maintaining long-distance connections in the brain.

The network uses leaky RNN units because of their ability to use activity from past time steps in a brain-like manner. Brain activity can persist for extended periods and so the model must reflect this. For each timestep, the hidden state of a unit is defined as

$$h_t = (1-\alpha)h_{t-1} + \alpha\tilde{h}_t,$$

where $\tilde{h}_t$ is the candidate hidden state given by activity and weight matrices for both input and recurrent activity, as well as any bias. The leak rate, α, determines the proportion by which the hidden state is updated and how much it 'remembers'. This was chosen over other RNN architectures, like LSTM and GRU, because it gives more continuous dynamics. Additionally, its simpler update rule requires fewer parameters and states variables. This therefore makes training faster and is more likely to be biologically realistic.

The stimuli used in each simulated task can be visual or somatosensory. Inputting these into the correct region of the brain is vital for the network to develop realistic solutions. Visual information must enter the network exclusively through the primary visual cortex, for example. In order to ensure this is the case, binary masks are applied to the input, defined as M_{vis} and M_{som}. Each mask is one-valued in only the rows of its respective sensory input and the columns representing its corresponding cortical region.

These masks are multiplied element-wise (Hadamard) to the weight matrix to give a masked weight matrix for each sense, denoted

$$\widetilde{W}_{\text{vis}} = W_{in} \circ M_{\text{vis}}, \quad \widetilde{W}_{\text{som}} = W_{in} \circ M_{\text{som}}.$$

To ensure sensory information does not have an effect on any other regions, the masked matrices are separately multiplied by the input vector, and the results are summed to give the final mapping of the input to cortical regions,

$$\mathbf{T}(t) = \widetilde{W}_{\text{vis}}\mathbf{x}(t) + \widetilde{W}_{\text{som}}\mathbf{x}(t).$$

$\mathbf{T}(t)$ represents the transformed input. It is the length of the hidden state, containing all cortical regions, and has non-zero values only in areas representing primary sensory regions. This method allows the use of a single input vector where segments represent different sensory information.

Within the hidden state, updates are controlled using the leaky RNN architecture discussed above, and the neural dynamics are given by

$$\tau \frac{d\mathbf{h}}{dt} = -\mathbf{h}(t) \ + \ \text{ReLU}\Big(W_{\text{rec}}\, \mathbf{h}(t) \ + \ \mathbf{T}(t) \ + \ \mathbf{b}_{\text{in}} \ + \ \xi(t)\Big),$$

where the neuronal time constant, τ, controls how quickly a neuron is updated $\mathbf{b}_{\text{in}}$ is the input bias vector of the network. Noise, denoted as $\xi(t)$, is also included in the hidden state and is sampled at each timestep from a normal distribution with mean zero and standard deviation σ, to be defined as a hyperparameter. This simulates the random firing of neurons seen in biological systems.

In the brain, once this sensory input is processed, a response to the task is given in the form of some kind of movement, usually from a signal from sent by the motor cortex. For the task paradigm used here, this response is a saccadic eye movement so the region output is read from is the frontal eye field (FEF). This is implemented in the same way as the input transformation, where a binary mask, M_{FEF} is applied to the output weight matrix to give a response vector,

$\hat{\mathbf{y}}(t)$, with zero-values everywhere except FEF, as given by

$$\widetilde{W}_{\text{out}} = W_{out} \circ M_{\text{FEF}}, \quad \hat{\mathbf{y}}(t) = \widetilde{W}_{\text{out}}\, \mathbf{h}(t).$$

2.2 Cost Function

The cost function is comprised of a core loss term based on task accuracy, along with further terms aiming to develop brain-like solutions. In order to train the network to successfully perform the tasks, the task loss, L_{task}, is weighted more heavily than the others. The network responds at every timestep, but the response which defines whether a trial is correct is at the end of the trial. The end of the trial therefore contributes more to the loss term, but the entire trial is included as the network must learn to fixate before giving a response. The mean squared error task loss is so defined as

$$L_{task} = \frac{1}{n}\sum_i \mathbf{m}\left(\hat{\mathbf{y}}_i - \mathbf{y}_i^*\right)^2,$$

where $\hat{\mathbf{y}}_i$ and $\mathbf{y}_i^*$ denote the prediction and target at output unit i, respectively. The total number of output units is defined as n. To emphasise the end of the trial, $\mathbf{m}$ is a non-negative masking vector which has higher values during the response period than during the fixation and stimulus periods.

A brain-shaped network alone is not sufficient to develop realistic connectivity. In order to encourage brain-like patterns, we can impose biologically inspired constraints and penalties during regularisation. There are many options here, representing different aspects of anatomy and energy conservation. Within the spatial regulariser, for example, it is not immediately clear whether distant connections should be penalised based on their absolute value, squared value or even exponentially as is suggested by some studies.[37] It was found that squaring the distance value multiplied by the weight, using L_2-style regularisation, gave the most realistic activity. While the brain does have sparsity, around two thirds of brain areas are connected, so using L_1-style regularisation may not be suitable here, as originally suggested.[4] By minimising the squared value, the network is pushed to create more small distance connections, giving a more continuous flow of activity through brain areas. This distance loss term is thus given as

$$L_{dist} = \sum_{i,j}(d_{ij} \cdot w_{ij})^2,$$

where d_{ij} and w_{ij} are the distance and weight between regions i and j, respectively.

Primary sensory areas are key to this model as they receive all the input. A problem identified in the model, however, is the over-use of these regions in higher cognitive processing. To avoid over-compression of the input vector, the nodes in sensory regions are replicated. The number of replicates can vary, with higher values facilitating easier training, but causing reliance on sensory areas. Too many strong recurrent connections were found in these regions, which is unrealistic and prevents activity spreading through the network into associative areas. A sensory activity penalty was implemented to prevent this over-reliance. An added loss term, L_{sens}, encourages the network to spread activity by penalising the total activity in primary sensory regions. This term is justified because in the brain, sensory information does not persist in primary cortices for very long, and complex cognition does not take place here. The loss term is calculated by multiplying the activity of the network element-wise by a binary mask selecting only primary sensory regions. The mean absolute activity over these regions can then be minimised in the cost function, and is defined by

$$L_{sens} = \frac{1}{n}\sum_{i=1} h_i m_i,$$

where h is the hidden layer activity vector, m is a binary mask with non-zero values in primary sensory regions and n is the total number of units representing these regions, including replicate units.

Another anatomical constraint tested was a dendritic spine count gradient. Dendritic spines are the sites of excitatory synaptic input into a neuron, and their number per neuron increases systematically as along the cortical hierarchy. A higher spine count may indicate more local or long-distance recurrent processing, which is thought to be a crucial property for higher cognitive function more cognitive function.[38] Incorporating this gradient could therefore be critical in reproducing realistic cognitive models of the whole cortex. Spine data for human brains is sparsely available, so to find a gradient, an estimated spine count was inferred using the inverse correlation between the ratio of a tissue property ratio, taken from structural imaging data, and the spine density.[39] To implement this into the model, a loss term, L_{spine}, is added to the cost function. This term increases when the sum of absolute connection values into an area varies from its estimated spine count. The estimated spine count data is scaled using the average input into a node, and the sum of regional absolute differences between the input connections and scaled spine count defines the loss, as given by

$$L_{spine} = \sum_i I_i - S_i, \quad I_i = \sum_j w_{ij}.$$

The total connection input into each region, I_i, is the sum of its row in the hidden layer weight matrix. The scaled estimated spine count for each region, denoted as S_i, can be subtracted from total inputs. Finally these values are summed for the overall spine loss.

With many possible loss terms and their associated weights, there is a large number of combinations for the cost function. Different combinations can be used to encourage the emergence of different brain-like characteristics. By finding the configurations which develop certain behaviours, we can draw conclusions about the causes of aspects of neurodevelopment. The three loss terms defined above were used with varying weightings for the results in this report, with a final cost function of

$$L_{total} = L_{task} + \lambda_1 L_{dist} + \lambda_2 L_{sens} + \lambda_3 L_{spine},$$

to be minimised in training using backpropagation, using a fixed learning rate α. Other biological constraints were considered and implemented, such as wiring entropy, but were not found to give insightful results for multisensory integration. Using more terms in the loss function can be detrimental to the explainability of the model, so should be avoided where they are not beneficial to dynamics or plausibility.

2.3 Task Paradigm

The tasks used for evaluating the network have been adapted from the 20-Cog-Task framework created by Yang et al. (2019).[8] They are proven to facilitate modularity and include a range of cognitive functions. For the two sense model, a subset of 9 tasks was used as this provided a good balance of variety and ability to train to a high performance. Developing a structure akin to the brain would require many tasks containing every type of cognitive process. This is not yet feasible for this network, however the subset of tasks comprises a range of unimodal and bimodal tasks, including those which engage different cognitive processes. For example, a visual delay-go task requires the use of working memory, as the network must remember the location of a stimulus. Other tasks require decision making or contextual integration to complete.

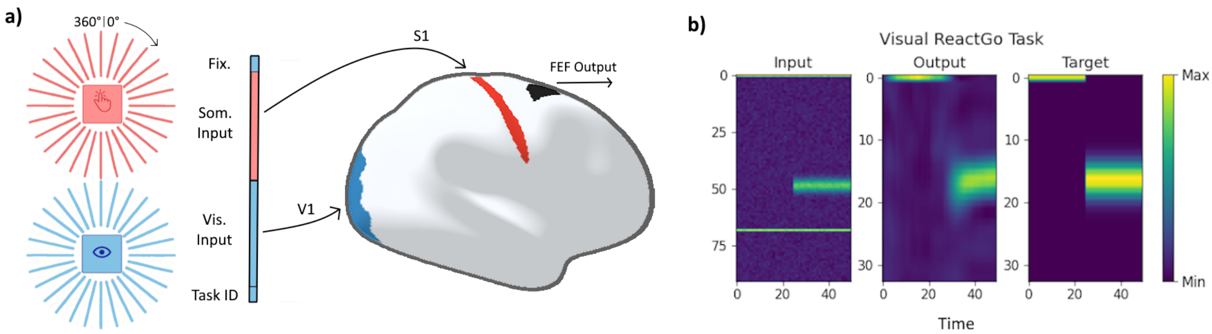

Figure 2: a) Simulated sensory locations are generated on rings and discretised into vectors with 32 units. In addition to the sensory information, visual task input includes fixation and task ID to inform the network when to respond and with what rule. Sensory information is mapped onto its respective primary sensory cortex, and output is read out from the frontal eye field (FEF). b) Simulated input and output is shown for a single trial of a visual react-go task, where an immediate saccade is made to the location of a stimulus. Network output is also shown, correctly responding to the task.

Sensory information of each modality is represented as a 32 unit vector corresponding to locations on a 360° ring. These two vector are concatenated and included in the input, as shown in Figure 2. Each 32 unit segment of the input enters the network only through its corresponding sensory region, as mentioned above.

Before any stimulus is shown, all tasks have a fixation period where the network must give an output of only the fixation, which is the first element of the input vector. This simulates the fixation stimulus focussed on in real cognitive testing. At the end of the task, when a response must be given, the network saccades to a given location. Because two tasks could have the same stimuli but different correct movements, contextual information must also be included in the input. At the end of the input vector, there is a binary block of nine elements corresponding to the nine tasks. The relevant task ID will be one-valued for the duration of the trial. An assumption of the model is that fixation and task ID are included in the visual information, which is likely to be the case in real testing.

Although an output is given at every timestep, the response period at the end is more heavily weighted in the cost function because task performance is most important. Early periods are still included in the cost function in order to reward focus on fixation, albeit to a lower degree.

2.4 Three-Sense Model

The CERNN framework can effectively model aspects of cognition with two senses, however it is not sufficient for multisensory integration. In order to capture the spatial dynamics, the auditory system cannot be overlooked. We constantly process auditory input in perception, so it is critical to include this in the model. Additionally, multisensory brain data is more readily available for auditory tasks than for somatosensory tasks, especially for audio-visual tasks. Therefore the network can be better evaluated if it is included.

To implement auditory information into CERNNs, we can include a third ring and transform it onto the primary auditory region, A1, as with vision and somatosensation. This will increase the length of the the input vector, and therefore be more expensive to train. With this inclusion, the updated transformed input becomes:

$$\mathbf{T}(t) = \widetilde{W}_{\text{vis}}\mathbf{x}(t) + \widetilde{W}_{\text{som}}\mathbf{x}(t) + \widetilde{W}_{\text{aud}}\mathbf{x}(t), \quad \widetilde{W}_{\text{aud}} = W_{in} \circ M_{\text{aud}},$$

In order to train all senses equally, the taskset must be updated. An auditory version of each task is added, and multisensory tasks are adapted to have a version for each combination of modality pairs. This gives an updated taskset of 15 tasks, consisting of 3 types of react tasks, as well as unisensory and multisensory decision making tasks.

2.5 Macaque CERNN

To verify results and draw conclusions about general cognitive mechanisms, CERNNs can be used for cross-species analysis. Non-human primates offer the best comparison to humans. Here a macaque parcellation is used, but the framework is flexible to any species where anatomical data is available. The parcellation used is based on work by Markov et al. (2014)[40] which has 91 regions. Forty of these regions have been further analysed using invasive viral injections in order to better map their connectivity.[6] For the best direct comparison with empirical data, this forty region subgraph is used for the CERNN. In surface plots the entire brain is shown, but some regions are not included in the network and will therefore stay zero-valued.

The auditory cortex is not included in the subset of regions, so only visual and somatosensory tasks are used for cross-species analysis between human and macaque CERNNs, within the scope of this report. Expansion to the entire brain is possible and only requires anatomical data. Primary and associative regions are generally organised similarly in both species. The visual cortex is located in the occipital lobe towards the back of the brain, and the somatosensory cortex is mostly in the parietal lobe, at the top of the brain.

The network trained using macaque data is configured in largely the same way as the human networks, however there is a difference regarding the distance penalty. Fewer long distance connections are found in humans when compared to macaques,[41] suggesting a stronger preference for localised subnetworks. Therefore a weaker spatial regulariser is used in macaque CERNNs, typically around an order of magnitude lower than for humans. Finally, spine count data is available for 27 regions,[42] and the remaining values are inferred based on the cortical hierarchy.[6]

3 Training and Evaluation

CERNNs are trained on all tasks in parallel, facilitating generalisable solutions and neural flexibility. Training continues until the average performance reaches 90%, with many configurations reaching this within 140 epochs. Models were evaluated based on task performance, alignment of connectivity with real data, and task-evoked activity patterns.

Although high accuracy is necessary, it is not what we are looking to for insights. The connectivity of the network and the flow of activity when performing tasks will inform us on the ways it replicates brain structure and function. The activity of the network is obtained by running a batch of test trials for a single task and averaging the hidden state over trials. While timing and strength of stimuli is random in training, these are fixed in test trials. This allows us to get meaningful results from averaging over a batch, which accounts for the random noise applied in hidden state updates. A brain state is given for each timestep, with a value for each cortical region. Along with the weight matrix, this can be used to evaluate the network.

To isolate the neural dynamics of multisensory integration and processing to complete the task, resting state activity must be removed. Before any stimuli are shown, the network is observed to reach a steady state by the end of the initial fixation period. For each task, this steady state can be found and removed from the entire activity array. This mirrors techniques used in fMRI imaging and allows us to see the direct response to the stimulus, and the processing used in reaching a response.

For reproducibility, Table 1 shows default hyperparameters. These are the values used for all models in this report unless stated otherwise. These values were usually chosen by performing parameter sweeps, although rather than choosing for pure performance, hyperparameters were generally chosen for the brain-like behaviour they cause.

Parameter Description	Symbol	Value
Distance regularisation weighting	λ_1	10^{-9}
Sensory regularisation weighting	λ_2	0.5
Spine regularisation weighting	λ_3	10^{-6}
Visual replicate number	n_{V1}	10
Somatosensory replicate number	n_{S1}	10
Auditory replicate number	n_{A1}	10
Motor replicate number	n_{FEF}	10
Learning rate	α	0.001
Timescale	τ	100
Standard deviation of noise	σ	0.01

Table 1: Network Hyperparameters

4 Results

4.1 Training on A Range of Tasks Leads to Brain-Like Connectivity

CERNNs are able to learn all tasks to a level comparable to or exceeding humans, with some configurations taking longer to reach this benchmark. The networks are very sensitive to changes in cost function hyperparameters. Specifically, increasing the distance penalty strength, λ_{dist}, slowed training significantly, and values exceeding 10^{-6} prevented average accuracy from reaching even 60%.

Models reaching performance of above 90% usually did so within around 80 training epochs of 200 steps each, as shown in Figure 3a. Unsuccessful models such as those mentioned above did not give any indication of reaching high accuracy when trained up to 150 epochs, suggesting that a stronger distance penalty is enough to make some tasks impossible for the network. Specifically, the network struggled with more complex including both unimodal and multisensory decision making.

While an increase in accuracy corresponds significantly to a decrease in loss, loss continues to drop once near perfect accuracy is reached. The task loss is dominant in the cost function, but other terms are also minimised. The continued decrease once accuracy converges indicates the structural optimisation of the network once it has learnt the tasks. This could be from solving the tasks with shorter connections, matching realistic spine count style inputs, or by doing less computation in primary cortices. These are all brain-like features, and so leaving the network to train after high performance is reached can be beneficial. Network dynamics in the later stages of training could make CERNNs a useful tool for exploring the evolution of brain-like connectivity under competing functional and structural constraints.

However, over-training can also be detrimental to results and analysis as the regularisation penalties can cause network weights and activity to be driven to very small values. This leads to the over-reliance of certain units, and is not a biologically-realistic outcome as strong signals are not necessarily discouraged in the brain. Moreover, over-optimisation to the point of convergence may not possible in brain due to well-documented problems with the biological plausibility of

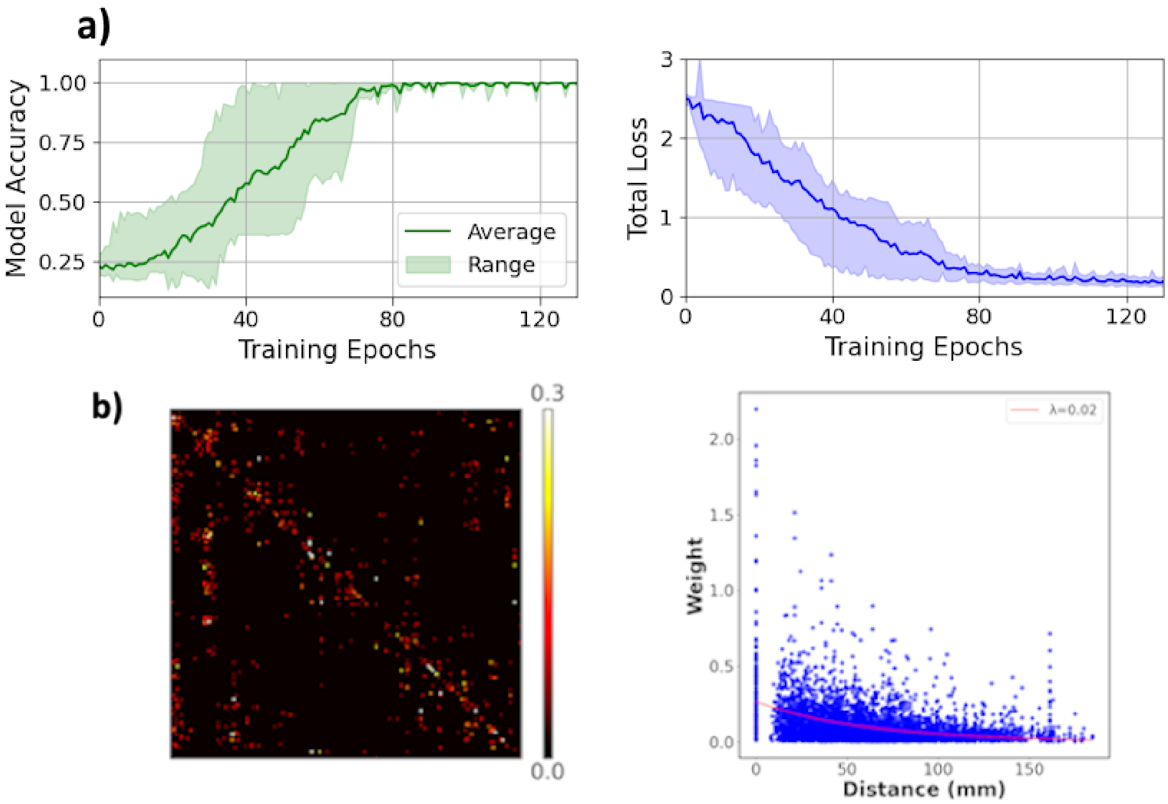

Figure 3: a) Average task accuracy and loss during training, taken over 10 trained networks. Accuracy is calculated here as the average over all tasks. b) The connectivity of the network is shown by a trained weight matrix and a scatter graph showing the distance distribution of weights. An exponential decay has been fit to the data to show the decrease in weight strength as units get further away.

backpropagation. To avoid this, network training can be stopped early. For example, when a wide spread of activity is targeted, a threshold of total weight values or activity can be included. Once the network has learnt the tasks, the model will cease to train if activity decreases to this threshold. Early stopping is a technique often used in machine learning to prevent overfitting,[43,44] although in this case the objective is different. While it is important to avoid overfitting in CERNNs, early stopping is used here to prevent a low activity solution, which is not usually an issue in machine learning given the output is correct.

The connectivity of the network can be compared to the human brain. Figure 3b shows the distance distribution and connectivity of weights in the network. Many self-connections are observed, which is realistic as most brain regions are heavily recurrent. This behaviour is expected to develop as self connections have a distance of zero, and so are not penalised in the spatial regularisation. The network is very sparse, which could be explained by the taskset as it does not necessarily require all aspects of human cognition. Biological systems must respond to far more than what is included in these tasks, and so they must be more flexible. The function of many connections in the brain is beyond the scope of these tasks, and so may not be represented in these networks.

The probability of a region being connected with another decreases exponentially in the brain as distance increases. Figure 3b shows an exponential curve fit to CERNN connectivity where this distribution can be approximated. Many of the strongest connections have low distances, perhaps mirroring the abundance of recurrent connections in the brain.

The sensitivity of CERNNs to changes in regularisation parameters suggests that cognitive function could be strongly linked with structural constraints. Specifically, increases in distance penalty prevented the learning of tasks which were otherwise relatively quick to train to near-perfection. This relationship could represent a trade-off in biological systems, where while minimising metabolic cost is beneficial, an excessive constraint on long-range connections would prevent connectivity necessary for complex higher cognition. Finding the right hyperparameters to balance this in CERNNs mirrors the biological compromise, where long-distance connections persist despite the metabolic cost. Additionally, the exponential decay in connectivity strength is notable as it aligns with empirical studies in anatomy. This could suggest that beyond capturing qualitative properties of the brain, CERNNs can quantitatively match key aspects of cortical organisation.

4.2 Complex Tasks Require More Cortical Regions

CERNNs learnt all tasks to a high degree of accuracy, but some tasks were considerably easier. Regardless of modality, simple tasks like react-go could usually be learnt and perfected within around 10 epochs. Even tasks like delay-go, which requires aspects of working memory, posed no problems for the network.

Decision making tasks were more difficult to learn, often converging at around 80% accuracy. Interestingly, while both unisensory and multisensory task versions struggled, the network had consistently lower accuracy for unisensory decision making tasks, as shown in Figure 4a. The set up of both of these tasks is identical, with the only difference being whether the two stimuli are presented on the same or different modalities. This suggests that it is easier to distinguish between senses than it is for a single sense. This can be expected in humans as it is thought that each type of sensory information is initially represented in different ways[1] and is therefore easier to discriminate from others. Also stimuli from the same sense are more likely to be confused with each other. This was not expected in the network, especially considering the distance loss term to make multisensory processing more difficult.

As well as having lower performance in decision making tasks, the activity of the network is also more distributed. More regions need to be recruited to solve more complex tasks. In Figure 4b, activity is shown for auditory decision making and react-go tasks. Stimulus and response periods have been averaged over in order to isolate cognitive processing. It is worth noting that the react-go task only has fixation and response periods, as a saccade is required immediately when stimuli are presented. The more complex task has activity in more than one area, specifically in parts of the prefrontal cortex (PFC). The primary sensory region here has less activity than the simpler task, suggesting the network is optimised such that complex cognitive function is carried out in areas higher in the cortical hierarchy.

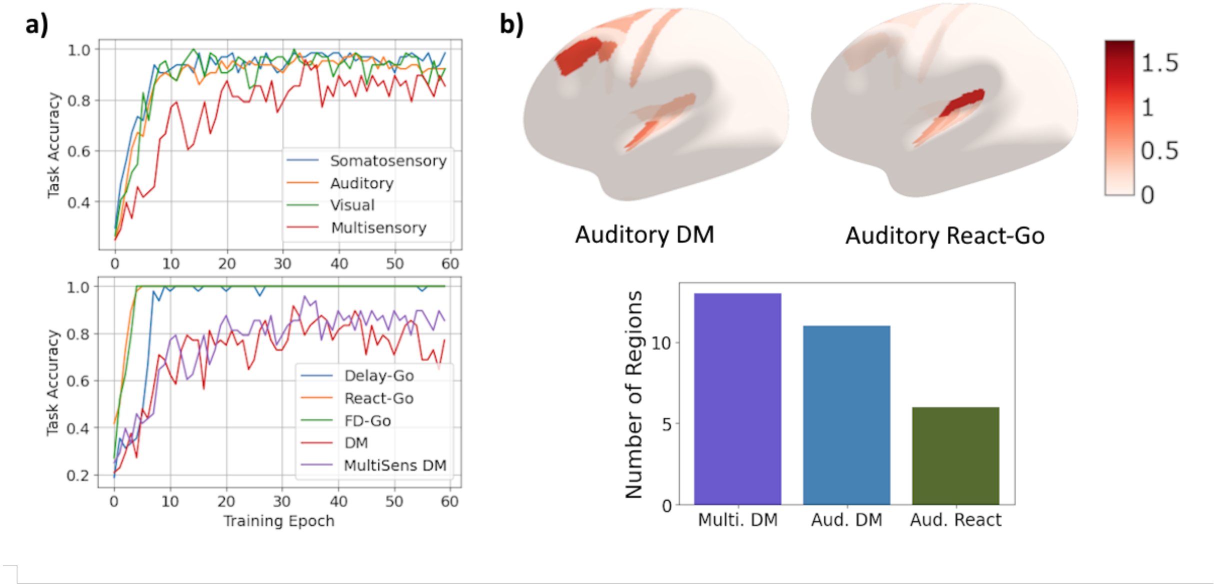

Figure 4: a) Model accuracy during training for tasks grouped by modality (top) and by task type (bottom). Each sense performed similarly, with multisensory tasks around 15% lower by the end of training. Decision making tasks have considerably lower accuracy than simpler tasks like react-go. b) Distribution of activity is shown for two auditory tasks. Activity is averaged over stimulus and response periods, to be plotted on the brain surface (top). Decision making tasks show more regions recruited. The number of active cortical regions, defined by an average activity of above 0.05, is shown for tasks varying in complexity (bottom).

Analysing task-specific activity distribution reveals insights into how CERNNs allocate computational resources based on task complexity. Simple tasks like react-go show very localised activity, perhaps suggesting a direct sensorimotor mapping , requiring minimal processing in order to complete the task. Direct connections like this are beneficial to both CERNNs and biological systems because activity takes time to propagate through the brain, and sensory updates sometimes demand instant action. Fast reaction time is an advantageous evolutionary trait, and avoiding unnecessary propagation to associative regions is likely developed to improve task efficacy, in this case being able to saccade within a single timestep once stimuli change. Importantly, the network seems to know when this kind of immediate response is needed, and when further processing is needed. In the latter case, activity is directed towards other cognitive regions as shown in Figure 4b.

Multisensory tasks engage the most cortical regions, as shown in Figure 4b. An increase in active regions corresponds to an increase in overall computational cost, suggesting multisensory tasks are more of a burden on the brain. CERNNs perform better at multisensory decision making tasks than unisensory equivalent tasks, suggesting multimodal information is beneficial for cognitive capability, despite the added burden. This demonstrates the use of having multiple senses in acquiring a general perception, particularly by aiding in discrimination of stimuli. Another biological compromise of the brain could be balancing the spatial and computational resources needed for multiple sophisticated sensory systems with the improved performance they offer.

Considering the number of regions engaged by different tasks can also begin to explain the corresponding propagation of activity. The difference between unisensory and multisensory decision making is relatively small, considering multisensory tasks have input in two different areas of the brain. This suggests that a shared processing region is developed, which both primary senses direct activity to. For the network, this minimises the number of connections, thereby optimising structure. An important development in neuroscience was the discovery of distributed cognitive networks.[31] Common regions and pathways found in CERNNs could begin to explain why and how these networks exist in the brain.

4.3 Activity Shows Realistic Representation of Working Memory

The main goal of CERNNs is to create computational models of the brain which solve tasks in realistic ways. Connectivity is critical and has been discussed above, but by analysing activity patterns, the mechanisms by which input is processed into a correct response can be better understood. Realistic activity was observed in several ways under different circumstances. Some behaviours are common to almost all successful models and some were specific and sensitive to configuration and hyperparameters.

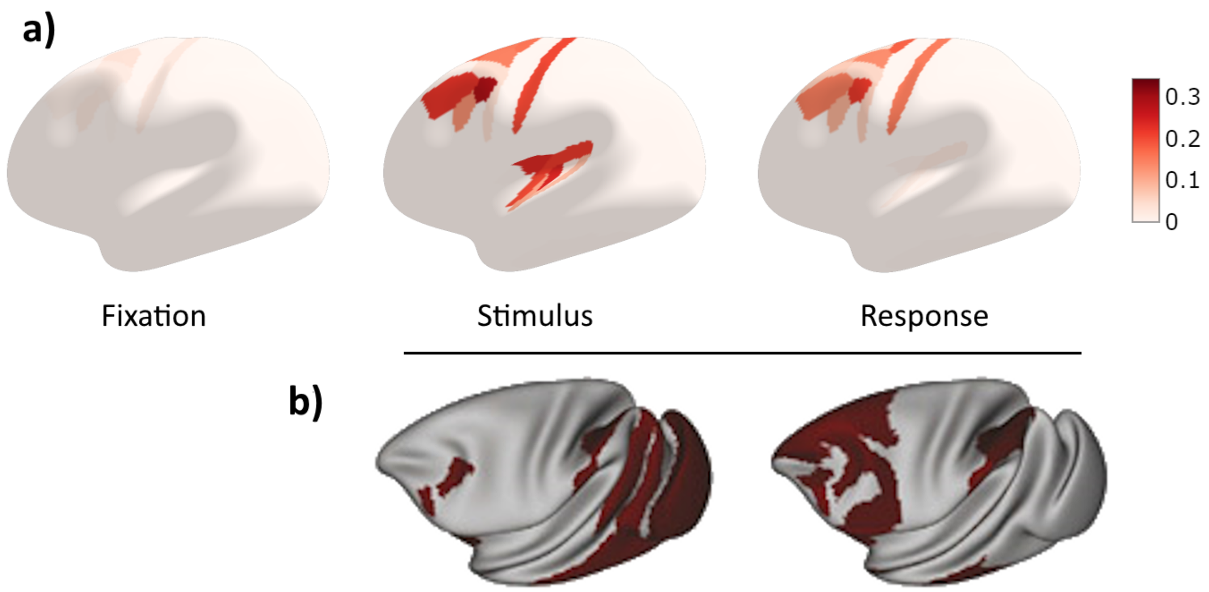

Figure 5: a) CERNN activity during a audio-somatosensory decision making task where the agent must saccade to the location of the stronger stimulus, after auditory and somatosensory stimuli are shown. Activity in the motor cortex and parts of the prefrontal cortex persists, while sensory regions are no longer active. Average resting state activity has been removed. b) Macaque fMRI brain activity during stimulus and delay stages of a visual working memory task, adapted from Froudist-Walsh et al. (2021)[6]

During the stimulus period, activity is prevalent in primary sensory regions and in parts of the prefrontal cortex and motor cortex. Following fixation and stimulus periods, successful trials show a sustained response. This is shown by persistent activity in FEF and often the wider motor cortex. Additionally, PFC regions stay active even after the sensory regions die out, as shown in Figure 5a. This suggests that some representation of the stimuli is still present in these regions, allowing the network to continue to respond correctly.

This activity mirrors the mechanisms in the brain by which sensory information is propagated towards the front of the brain, up the cortical hierarchy, to be stored in the working memory. Storing information like this is required for many tasks, but cannot prevent new information being received into the brain. Working memory therefore must not be stored in sensory regions. Primary sensory areas act transitively in this process, while representations are sustained when necessary in the PFC.[6,38] These regions stay active even when no new input is received and the response is unchanged. Recurrent connections are used to keep a memory of the task, perhaps necessary to sustain the correct response in the output nodes.

To validate the biological realism of CERNN activity, a qualitative comparison was made with empirical activity. Non-human primate (macaque) fMRI data[6] for working memory tasks shows similarities between biological systems and CERNNs. Figure 5b shows activity for stimulus and delay periods, where a monkey must remember the location of a stimulus after it has disappeared. While a response is not made in these periods of the experimental task, a comparison can still be made as the delay requires sustained representation of the task, much like the simulated CERNN tasks. It is important to note that the stimulus was not shown for a sustained period, which explains why the stimulus period has no activity in associative areas, in contrast to CERNNs. The information is yet to propagate forward. Sensory information is initially almost exclusively represented in sensory regions, towards the back of the brain. During the delay however, the representation has shifted and is sustained in the prefrontal cortex, much like CERNN activity. Although difference in task design and species must be acknowledged, the model's correspondence with empirical data strengthens the reliability of solutions.

These activity patterns show that not only do CERNNs exhibit brain-like properties, as shown by a region of persistent activity for working memory tasks, they also provide anatomically plausible solutions. Previous neural network models may develop a similar cluster for storing relevant information, but without anatomical data this cannot emerge with physical relevance, and therefore real-world application. The spatial element gives CERNNs an extra dimension of explanation and insight in understanding the functional organisation of the brain. Localisation of sustained activity to the motor cortex and parts of the PFC reinforces the model's capability to replicate real cortical dynamics without explicit programming of functional roles. The location of this working memory representation is not trivial, as there is no bias in the network to cause it to develop there. It can be only a direct result of the biological constraints, and the minimisation of distant connections. From this, predictions can be made about the spatial relationship between sensory, associative and motor regions in the brain.

4.4 A Common Region of Multisensory Integration Emerges

The use of early stopping with a smaller distance regularisation parameter leads to a wider spread of activity. This is useful because the cognitive processes are often shown more clearly before the network over-optimises, as detailed previously. These clear dynamics can be used in trying to isolate the specific mechanisms of multisensory integration. Analysing activity for decision making tasks between each pair can help to identify the relevant regions or networks.

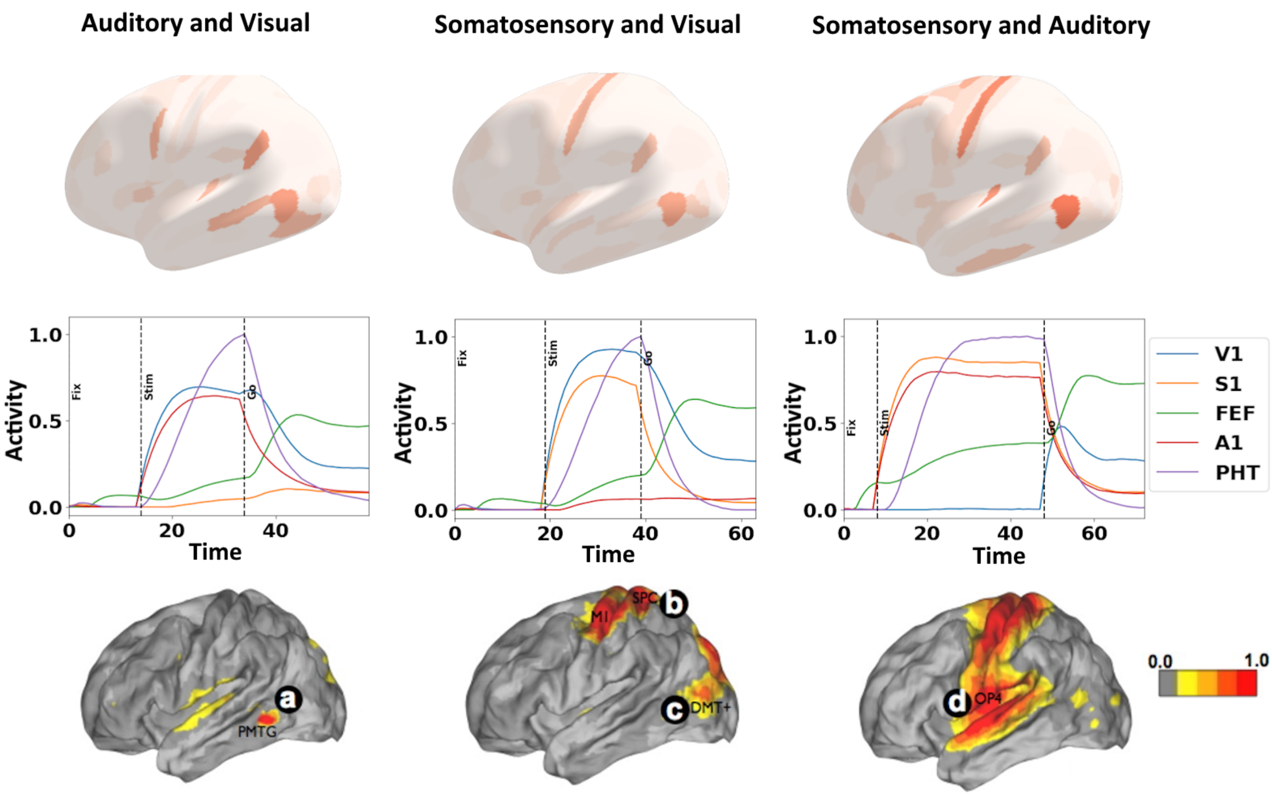

Figure 6: Average network activity during stimulus periods of multisensory decision making tasks. A weaker value of $\lambda_{dist} = 10^{-8}$ was used, and training was stopped after accuracy reached 90% at around 40 epochs. Surface plots of average stimulus period activity show shared regions across modality pairings, as well unique characteristics for each (top). Plotting normalised activity over time for key regions shows region PHT receives large amounts of activity shortly after sensory regions, suggesting it is a used in propagating sensory information (middle). Adapted from Sepulcre et al. (2012),[45] bimodal integration regions are shown for each modality pair, found using stepwise functional connectivity analysis of resting state fMRI data (bottom). Bimodal integration was most prominent in regions labelled, specifically between auditory and visual (a), somatosensory and visual (b, c) and somatosensory and auditory (d) modalities.

Surface plots of the brain give indications on the location of each type of multisensory integration. For each pairing, activity in the corresponding sensory regions is strong, along with some weak activity in and around the motor cortex, although this is stronger in the audio-somatosensory case. Beyond these expected regions, there are two additional regions which consistently show considerable activity: area PHT in the lower right of the surface plots and the Perisylvian Language area (PSL), found in the middle right of the plots. PHT was the more prominent of the two in all three cases, but it seems that both are important in integrating sensory information. They are both considered to be in the region of the temporo-parietal junction, which is thought to be a primary site for all kinds of multisensory integration.[20,21]

In addition to these key shared areas, each modality pair displays some unique regions likely used in integration or processing. For auditory-visual tasks activity bridges from A1 to V1, engaging area PHT and other neighbouring regions in the occipital and temporal lobes. This activity is consistent with anatomical locations indicated by much research,[20,21] suggesting that CERNNs are especially accurate spatially for this modality pair. The other pairs are less directly comparable, with visual-somatosensory tasks having low activity in almost all outer brain regions and auditory-somatosensory having a wider spread of distributed activity unlikely to be biologically realistic.

Further analysis of common regions gives insight into their role. PHT in particular appears to be a crucial for effective and efficient integration. Plotting the temporal evolution of key areas, as in Figure 6 (middle row), shows how and when each region is most used, helping to clarify the dynamics of integration. For all three pairs, activity in PHT rises significantly shortly after sensory regions spike corresponding to stimulus onset. This suggests the existence of direct connections from primary sensory regions to PHT, indicating it could act as a critical relay or integration node regardless of the modalities involved. It is likely that there is some unimodal processing involved in primary sensory cortices before integration, perhaps corresponding to the delay in activity between sensory regions and PHT. Following this, activity propagates forward to PFC areas for further processing or directly to FEF to give a response. One the region is identified, incoming and outgoing connections to PHT can be analysed. It was found that connections to PHT from primary sensory regions were an average of 36% stronger than to other areas. Similarly, connections from PHT to FEF and the wider motor cortex were 25% stronger than to other regions. This evidence strengthens the idea that PHT works as in intermediate integrative hub, where sensory information is processed before being used in responding to tasks.

Comparison to stepwise functional connectivity analysis from resting-state fMRI, adapted from Sepulcre et al. (2012)[45] provides further validation of these findings. Figure 6 (bottom row) shows the regions of bimodal integration found for each pairing. Again, auditory-visual CERNN activity shows the most correspondence with empirical data, with an almost direct overlap around the posterior temporal region. Visual-somatosensory and visual-auditory activity do also show some similarities, particularly surrounding primary sensory regions. These patterns emerge without explicit bimodal labels, suggesting that the evolution of the brain's integration system is optimal under the biological constraints included in the model. The flow of information which develops from primary cortices into integrative regions before propagating forward to associative areas aligns conceptually with theories of hierarchical processing in the brain. Multisensory information is first transformed and integrated locally and then abstracted through as it moves into higher-order areas. This suggests that CERNNs not only succeed in task performance but also develop a functional organisation that can change depending on task demands, reflecting the flexibility of biological systems.

4.5 Cross-Species Verification Using Macaque CERNN

CERNNs have many shared properties with the brain. Thus far, human anatomy has been used to find these properties, however by extending the model to different species we can verify these findings and identify the fundamental mechanisms behind multisensory integration and other cognitive processes. Two models were trained using nine somatosensory and visual tasks. For the human model, a distance regularisation parameter of $\lambda_{dist} = 10^{-8}$ was used, whereas a weaker value of 10^{-9} was used for the macaque model, aligning with empirical data.[41] All other parameters were kept the same in order to make a direct comparison of learning dynamics and functional organisation between species. Both models avoided over-compression by replicating input and output regions to have 10 units each.

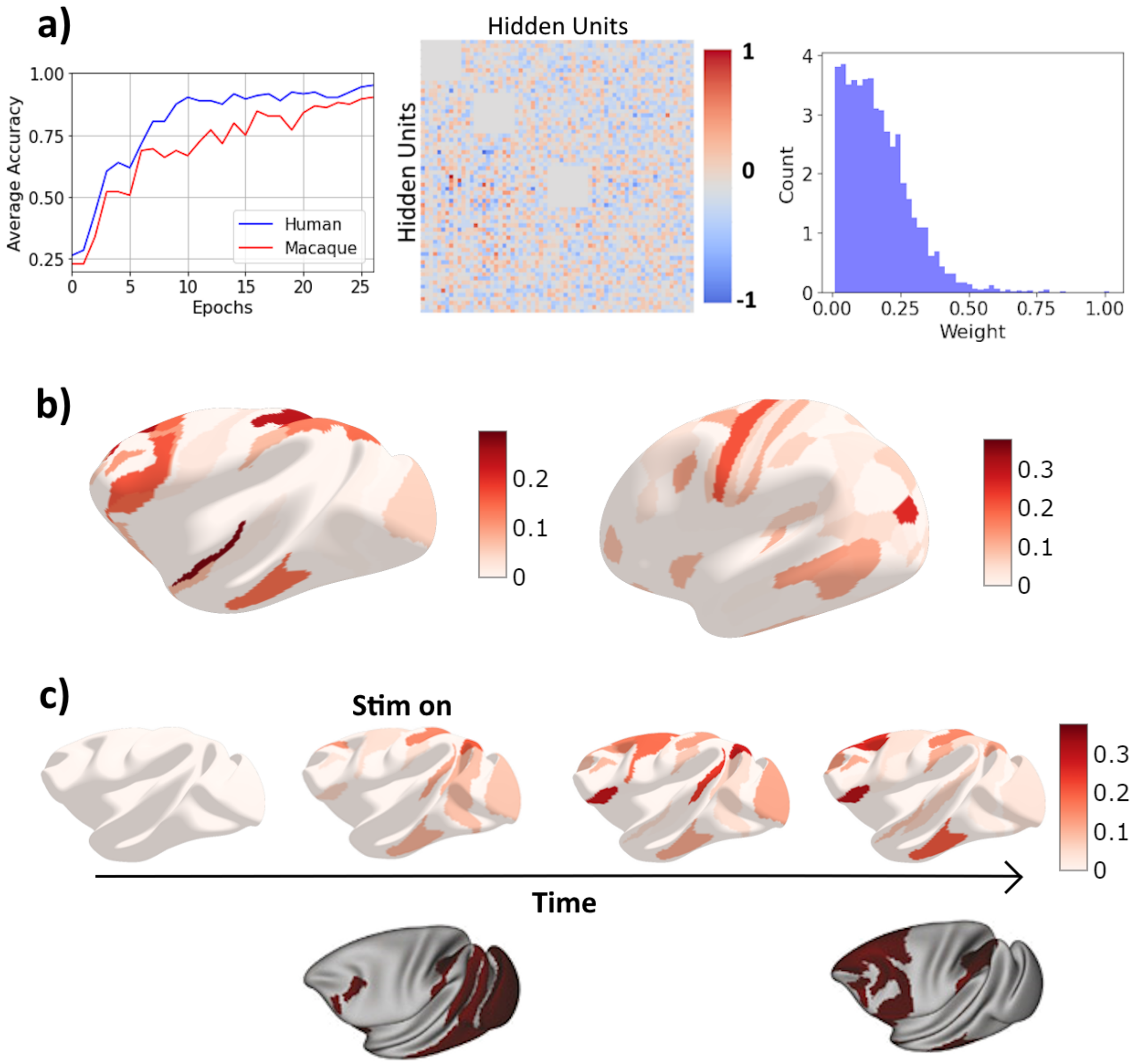

Figure 7: Macaque and human bimodal models were trained on 9 tasks. a) Training accuracy for humans outperforms macaque, however both species reach 90% accuracy within 25 epochs. Trained weights show moderate sparsity other than in replicated regions where recurrent connections are penalised. The distribution of weights shows many weak connections. b) Surface plots for response period in a somatosensory-visual decision making task. Somatosensory and motor regions are highly active in both, as well as parts of the temporal and occipital lobes. c) Temporal analysis shows the propagation of information from visual to motor cortices during a react-go task. Comparison with a similar task is made using fMRI data.[6]

Both models were similarly capable at completing the tasks although the human model generally outperformed the macaque model, learning tasks faster and to a higher accuracy, as shown in Figure 7a. This could be due to the increased size of the brain, and therefore network, in humans. It is important to note that only a subset of cortical regions in the macaque were used, limiting computational power. Average accuracy reached 90% within 25 epochs for both models. This is faster than many models discussed in this report, likely explained by the reduced size and complexity of the taskset with only two modalities. While the structural differences between species affects learning, both architectures converge to capable cognitive networks.

The weight matrix of the network shows it has less sparsity than the human CERNN. Most regions are connected, although connections within replicates blocks are penalised in order to promote a spread of activity. Most weight values are low valued, as also shown by the histogram in Figure 7a. A positive skew indicates a small number of strong connections. This suggests the network recruits many regions to solve tasks, preferring to spread activity around the brain rather than relying on well defined pathways to guide lots of activity into specific strongly connected areas. Less sparsity could be explained by considering brain size and cognitive demand. A higher proportion of possible connections are used in the macaque, perhaps suggesting that higher cognitive processes require more brain. This could even be extended, indicating that humans are capable of more advanced cognition because only a subsection of their brain is required for the same tasks.

To better understand the relationship between species, neural activity can be observed when completing the same task. Figure 7b shows surface plots of activity averaged over stimulus and response periods of a somatosensory-visual decision making task. Both networks show strong activity in the motor regions used for output, which is closer to the front of the brain in the macaque model than the human. Activity is largely biologically plausible, with somatosensory and visual regions showing activity, as well as the surrounding areas. Additionally, both networks have areas active in the temporal lobe, or around the temporo-parietal junction. As previously discussed, this could indicate a region of sensory integration, although for the macaque in particular the location of this is not expected considering the modality pairing. The existence of these regions suggests that information could be propagated from sensory regions into an intermediate region before being sent forward to the motor cortex. If found to be the case, the coexistence of this mechanism (initial sensory activation, posterior integration, and anterior output) in both macaque and human CERNNs would indicate that it is optimal under the included constraints, and therefore give explanation to the specific factors leading to cortical organisation in biological systems.

Temporal macaque analysis gives remarkable brain-like behaviour. For tasks like react-go, the propagation of information can be seen to move from sensory regions towards the motor cortex, travelling with almost wave-like properties. For this task there is no delay; a saccade should be made immediately on stimulus onset. As soon as visual stimulus is presented, it is clearly observed in the occipital and parietal lobes. As time progresses, activity moves forward, recruiting subsequent regions as it travels up the hierarchy before finally culminating in the motor cortex. This can be seen more clearly in videos attached in Appendix A. By this point, activity in the visual cortex has begun to fade, but is sustained in the anterior of the brain, where recurrent connections and dendritic spine counts are more abundant in biological systems. Further comparison can be made with macaque fMRI data,[6] which shows corresponding sequential activation of the occipital, parietal and frontal lobes when completing a similar task. CERNNs therefore capture not only steady-state activation patterns, but also the biologically-plausible dynamic propagation of information across the brain. This emerges without needing to explicitly specify timing rules or labels, suggesting that the architecture and constraints in the model are sufficient to replicate realistic temporal processing. During the final period, parts of the temporal and parietal lobes are active in both CERNN and fMRI, but are not expected based on current understanding. The fact that these regions appear in CERNN activity as well suggests that they are used in completing the task or facilitating flexibility. This kind of finding could lead to predictions about the specific roles of units which previously did not seem important.

These results demonstrate that CERNNs can be adapted to different species without losing the core ability to learn tasks and develop brain-like features. Using largely the same constraints, macaque models exhibit similar behaviour to human models, even going beyond in showing more clearly defined temporal dynamics. CERNNs can be used as powerful tools in cross-species analysis of cognitive systems, exploring the relationship between all levels of neural processing and anatomy.

5 Discussion

The CERNN framework introduced in this report could offer a meaningful advancement in computational neuroscience by bridging the gap between highly capable artificial neural network models and the cortex-wide anatomical models, incorporating the data required to generate impactful insights. By integrating biologically-inspired constraints, this new architecture can begin to have naturally emerging cognitive behaviour.

Multisensory integration is an area of cognition which lends itself to CERNNs. There are known fixed input locations for each modality, which must be integrated somehow in order to solve complex problems. By incorporating the somatosensory, visual and auditory systems into the model, we can observe the optimised flow of activity under different constraints and conditions. A distance penalty representing wiring cost in the brain is likely to be a critical factor in its organisation, and so should be weighted accordingly. Other terms should be included however, for example a sensory activity regulariser was particularly necessary in this report.

CERNNs used primary sensory areas as transitive regions. Input is passed on to higher cognitive areas, thereby freeing up early regions for new and potentially salient stimuli. Once information reached higher regions, the network decided whether further processing was needed to complete the task, or perhaps this representation needed to be sustained in order to respond after a delay. By using the task context, CERNNs could effectively use the correct cognitive mechanism, and this usually took place in the prefrontal cortex, as in the brain. The networks develop a general region or network for higher cognitive processing close to the motor cortex, where output is read.

Results like this could guide hypotheses about the spatial relationship between cortical regions and how they have developed.

Beyond general principles which could have emerged in abstract RNNs, the ability of CERNNs to give accurate anatomical predictions is novel and impactful. Using the prefrontal cortex for sustained representations and the temporo-parietal junction for intermediate sensory integration prove the natural emergence of fundamental cognitive mechanisms. Temporal analysis, particularly in macaque models, showed a realistic flow of information. A tool with both of these capabilities can have great utility considering the trade-off between spatial and temporal accuracy in neuroimaging technologies.

To avoid over-compression of information, input and output nodes were replicated. This led to numerous issues to be solved by further constraints. Over-constraining a model does not give us useful insights, so to avoid these problems further work could encode information more concisely, or use a parcellation with finer resolution where multiple neighbouring regions are used for input. It is important to note that such a parcellation would increase the size of the network and therefore increase computational load.

In the current model, each type of sensory input is encoded in the same way, with the only difference being its input region. This is unlikely to be the case in biological systems where the structure and function of sensory systems varies greatly from initial receptors to cortical regions. This could limit the extent to which CERNNs can accurately model the details of multisensory integration. Distinct sensory encoding could be necessary to develop the true mechanisms of merging different modalities into a single stream of information. Specialised unisensory models could process input before entering the CERNN primary regions, perhaps replicating the propagation of signals from the eyes, ears and nervous system before reaching the brain. For example, visual input could be passed through a convolutional neural network (CNN) before entering V1. Note that deep CNNs should be avoided in order to not over-process data before entering the CERNN; excessive pre-processing will limit the use of CERNNs as brain regions may no longer be required for cognitive processing.

CERNNs are adaptable to any brain mapping. For any parcellation, distance between regions can be computed and used in the model. As can be seen from surface plots, using parcellations with a small number of regions, which are consequently large, can cause connectivity to be disjointed. A single distance value between large irregular parcels can be misleading. The Glasser parcellation is used here for its functional relevance, however for a more continuous analysis of the dynamics, a parcellation with more regions could be chosen, as well as a smaller timescale. This could show a clearer flow of information through many smaller regions.

Many interesting and some unexpected brain-like behaviours emerged in the network once trained, however this was often sensitive to hyperparameters and with the current setup it seems unlikely that a network would exhibit all these characteristics at the same time. While this could be a case of needing further tuning, it is likely that a deeper change in task paradigm and network architecture is required to create general brain-like networks. Rather than being a comprehensive and general model of the brain, CERNNs have been found here to be a useful tool in understanding specific mechanisms. Tuning for brain-like properties and analysing the required biologically-inspired constraints furthers understanding about both these properties and the reasons they have developed in the brain.

An important avenue for further investigation is the extent to which CERNNs can generalise when presented with unseen tasks or novel combinations of sensory modalities. The current model is trained on a fixed set of tasks, each associated with a predefined mapping from input to output. However, cognitive flexibility is one of the defining features of biological intelligence, and the ability to perform new tasks by recombining previously learned elements is critical for general cognition. In the brain, this is achieved through compositional representations and shared processing circuits, particularly in transmodal regions like the prefrontal cortex. CERNNs are suitable for this analysis as a result of their anatomical structure and multi-task training regime. Future work could involve training the network on a subset of tasks and then evaluating its performance on unseen task variants, such as novel modality pairings or altered task rules. If successful in these transfer learning scenarios, CERNNs would provide strong evidence for emergent generalisation, proving the models as not only biologically plausible but also functionally powerful. This would bring the model closer to the ultimate goal of CERNNs, a general-purpose cognitive architecture capable of flexible adaptation when posed with new demands.

Many more types of neuroscience data should be explored using CERNNs. For example, cross-species analysis could be insightful in finding general cognitive mechanisms and could be impactful in guiding experiments. Macaque models have been discussed here but by extending to more species such as mice, processes could be identified as general or species-specific. The ethical implications of this could this considered by neuroscientists when selecting species for experiments.

Another limitation of current CERNN models comes from the calculated distance between cortical regions. geodesic distance is used, where connections travel along the surface of the brain. In reality the brain is not a hollow shell, and many regions are highly connected inside the brain rather than just on the surface. These connections often pass through the deeper white matter beneath the cortex. A more realistic measure of distance could account for this by incorporate a combination of the two, perhaps using a weighted average between Euclidean and geodesic distances. Furthermore, as scanning technology improves and high resolution data becomes more available, distances could even be found empirically. Incorporating anatomical data is more likely to replicate brain-like processing.

If a successful general model of the brain can be reached, then it could be lesioned or constrained to investigate the effects of simulated neurodegenerative disease or traumatic brain injury. Regions of the network could be disabled to mimic these effects, and the adaptation of the network could guide treatment or experiments. Similarly, neurodegenerative diseases could be modelled by changes in timescale, acting as a decrease in memory capacity. The practical use of these models could be increasingly impactful as cases of brain disease rise with an aging population.

Beyond just modelling brain disease, CERNNs have potential to be applied in the simulation of treatment. Methods involving stimulation, such as electroconvulsive therapy (ECT) and deep brain stimulation (DBS) could be modelled by a burst of input into specific regions. The reaction and adaptation of the network under these conditions could inform further development and refinement of these treatments. The effects of medication could also be modelled using CERNNs. Many drugs work by altering the amount and balance of neurotransmitters across different cortical regions. As an example, antipsychotics lower the availability of dopamine to control hallucinations. A future version of CERNNs could accommodate for this, perhaps incorporating a dopamine gradient and suppressing activity in abundant regions. Certain tasks and cognitive mechanisms could be found to be affected in particular, and this analysis could even help guide the correct course of treatment at an individual level. The CERNN framework has been created to be applicable to both fundamental scientific and practical applications.

6 Conclusion

The development of the brain is influenced by many functional and biological factors. As constraints were added, refined and tuned, the model became more brain-like. Many core features of multisensory integration and processing, both spatial and temporal, could be captured by CERNNs. Analysing activity showed remarkable similarities with biological systems, and a brain-like network emerged after a relatively low amount of constraining. This development could lead to more accurate and directly testable predictions in understanding the mechanisms used by multimodal biological systems. These models expand on current state-of-the-art models by embedding real-world relevance in the form of anatomy. This makes them directly comparable to empirical neuroscience data across the brain, while maintaining the capability for difficult tasks which yield rich cognitive dynamics. Cross-species analysis further deepens and widens the use-case of CERNNs due to their flexible framework.

Although a CERNNs effectively modelled multisensory integration, particularly in humans, a general cortex-wide model which simultaneously exhibits a range of biologically-realistic behaviour would likely require more constraints. It could also improve the accuracy of simulated activity, leading to better alignment with fMRI activity and more reliable predictions. Therefore further implementation of neuroscience data will yield a more comprehensive simulation of cognitive mechanisms and should be explored in further studies. Neuroscientists can use this framework as a tool to improve the precision of model-guided experiments, a direct result of integrating anatomical data.

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A Videos of Macaque Activity

To illustrate the temporal dynamics of the macaque CERNN, videos have been created and can be accessed using the url:
`https://drive.google.com/drive/folders/1_x74oZrymOzxihc4sJ0JMfEtgWZP-FuC?usp=drive_link`